

Bank Marketing Campaign

Funnel & Conversion Performance Analysis

Data Analytics Internship — Task 3

Dataset: Portuguese Bank Telemarketing | Records: 45,211 | Period: May 2008 – Nov 2010

1. Executive Overview

This report analyzes the marketing funnel performance of a Portuguese banking institution's direct telemarketing campaign. The campaign aimed to sell term deposit subscriptions to customers through phone calls. The analysis covers 45,211 customer records and identifies key conversion bottlenecks, drop-off points, and actionable strategies to improve campaign performance.

Metric	Value
Total Customers Contacted	45,211
Total Subscribed	5,289
Overall Conversion Rate	11.70%
Average Call Duration	4.3 minutes (258 seconds)

2. Conversion Rates at Each Funnel Stage

The marketing funnel tracks customers from initial contact through to final subscription. Each stage represents a filtering point where customers either progress or drop off.

Funnel Stage	Customers	% of Total	Stage Conversion
Stage 1 — Total Customers Contacted	45,211	100%	Baseline
Stage 2 — Reached via Cellular	29,285	64.8%	64.8% reached
Stage 3 — Previously Contacted	8,257	18.3%	18.3% retained
Stage 4 — Subscribed	5,289	11.70%	Final conversion

Key Insight: Out of every 100 customers contacted, only approximately 12 actually subscribed. The biggest opportunity lies in improving Stage 1 to Stage 2 (contact method) and reducing drop-off at Stage 2 to Stage 3 (prior relationship).

3. Key Drop-off Insights

Transition	Customers Lost	Drop-off %	Root Cause
Stage 1 to Stage 2	15,926	35.2%	Unknown contact method used
Stage 2 to Stage 3	21,028	71.8%	No prior relationship with customer
Stage 3 to Stage 4	2,968	35.9%	Over-calling reduces interest

Drop-off 1 — Stage 1 to Stage 2 (35.2% Lost)

- 15,926 customers were lost because they were contacted via unknown or low-quality methods.
- 13,020 customers (28.8%) were contacted via an unknown channel — only 4.1% of them converted.
- Cellular contact converts at 14.9% vs unknown at just 4.1% — a 3.6x difference.
- Switching to cellular contact can recover a significant portion of these lost customers.

Drop-off 2 — Stage 2 to Stage 3 (71.8% Lost) — Biggest Drop-off

This is the single largest drop-off in the entire funnel with 21,028 customers lost.

- 36,959 customers (81.8%) had never been contacted before — new customers convert at only 9.2%.
- Previously successful customers convert at 64.7% — that is 7x higher than new customers.
- The bank is over-investing in cold outreach while under-utilizing its existing warm customer base.
- No prior relationship means no trust, which results in no subscription.

Drop-off 3 — Stage 3 to Stage 4 (35.9% Lost)

- Customers called 6 or more times convert at only 5.8% vs 14.6% for a single call.
- Over-calling is actively damaging the conversion rate.
- 2,968 customers dropped off at the final stage — many due to call fatigue.
- Setting a maximum of 3 calls per customer could significantly reduce this drop-off.

4. Performance Breakdown by Segment

4.1 By Contact Type

Contact Method	Total Customers	Subscribed	Conversion Rate
Cellular	29,285	4,369	14.9%
Telephone	2,906	390	13.4%
Unknown	13,020	530	4.1%

4.2 By Month

Month	Total Contacted	Subscribed	Conversion Rate
March	477	248	52.0%
December	214	100	46.7%
September	579	269	46.5%
October	738	323	43.8%
April	2,932	577	19.7%
February	2,649	441	16.6%
August	6,247	688	11.0%
November	3,970	403	10.2%
June	5,341	546	10.2%
January	1,403	142	10.1%
July	6,895	627	9.1%
May	13,766	925	6.7%

4.3 By Job Type

Job Type	Total	Subscribed	Conversion Rate
Student	938	269	28.7%
Retired	2,264	516	22.8%
Unemployed	1,303	202	15.5%
Management	9,458	1,301	13.8%
Admin	5,171	631	12.2%
Self-Employed	1,579	187	11.8%
Technician	7,597	840	11.1%
Services	4,154	369	8.9%
Housemaid	1,240	109	8.8%
Entrepreneur	1,487	123	8.3%
Blue-collar	9,732	708	7.3%

4.4 By Education Level

Education Level	Conversion Rate
Tertiary	15.0%
Unknown	13.6%
Secondary	10.6%
Primary	8.6%

4.5 By Previous Campaign Outcome

Previous Outcome	Total	Subscribed	Conversion Rate
Success	1,511	978	64.7%
Other	1,840	307	16.7%
Failure	4,901	618	12.6%
Unknown / New	36,959	3,386	9.2%

4.6 By Number of Calls Made

Calls Made	Total Customers	Conversion Rate
1 Call	17,544	14.6%
2 to 3 Calls	18,026	11.2%
4 to 5 Calls	5,286	8.6%
6+ Calls	4,355	5.8%

5. Actionable Recommendations

Recommendation 1 — Switch All Contacts to Cellular

Impact: Very High | Effort: Low | Priority: #1

- Cellular converts at 14.9% vs unknown at 4.1% — a 3.6x improvement.
- 13,020 customers were contacted via unknown method — shift all of these to cellular immediately.
- Potential gain: approximately 1,400 additional subscriptions from this one change alone.

Recommendation 2 — Re-target Previously Successful Customers First

Impact: Very High | Effort: Low | Priority: #2

- Previously successful customers convert at 64.7% — 7x higher than new customers at 9.2%.

- Build a priority call list of the 1,511 past successful customers before any cold outreach.
- These customers already trust the bank — they need less convincing and fewer calls.
- This is the single highest ROI action available to the campaign team.

Recommendation 3 — Focus Campaigns in March, September, October and December

Impact: High | Effort: Low | Priority: #3

- March alone converts at 52.0% — nearly 5x the overall average of 11.7%.
- September (46.5%), October (43.8%), and December (46.7%) also show very strong performance.
- May is the worst month at 6.7% — reallocate May budget to high-performing months.
- Concentrate 60 to 70% of campaign resources in March, September, October, and December.

Recommendation 4 — Set a Maximum of 3 Calls Per Customer

Impact: Medium | Effort: Low | Priority: #4

- 1 call converts at 14.6% — 6 or more calls drops to only 5.8%.
- Over-calling actively damages the brand and wastes agent time.
- After 3 failed attempts, move customers to a low-priority re-engagement list.
- Redirect saved agent time to high-priority warm leads.

Recommendation 5 — Prioritize Students and Retired Customers

Impact: Medium | Effort: Medium | Priority: #5

- Students convert at 28.7% and retired customers at 22.8% — both are 2 to 3x the average.
- Design specific campaign scripts tailored to these two high-converting segments.
- Offer relevant incentives such as savings goals for students and pension benefits for retired customers.

Recommendation 6 — Improve New Customer Onboarding

Impact: High | Effort: High | Priority: #6

- New customers make up 81.8% of all contacts but convert at only 9.2% — massive room to improve.
- Introduce a better pitch script with a clear value proposition for first-time contacts.
- Add follow-up SMS or email after the initial call to nurture interest.
- Improving the new customer rate from 9.2% to 12% would add approximately 1,000 more subscriptions.

6. Summary — Top 3 Quick Wins

Priority	Action	Expected Impact
1	Re-target previously successful customers first	64.7% conversion rate — 7x higher ROI
2	Switch all contacts to cellular method	3.6x improvement over unknown channel
3	Run heavy campaigns in March and Q4 months	52% conversion rate vs 11.7% average

Final Note: The core insight of this analysis is that the bank's marketing success depends less on volume and more on targeting the right customers, at the right time, through the right channel. Quality over quantity is the key to improving conversions.